The empirical recurrence for OEIS A240297, recovered from its tail

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Abstract

OEIS A240297 records "Empirical recurrence of order 80" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 125$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{80} = 8508416 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A240297 is "Number of 3Xn 0..3 arrays with no element equal to zero plus the sum of elements to its left or two plus the sum of the elements above it or two plus the sum of the elements diagonally to its northwest, modulo 4.". It has offset 1 and begins

$$4, 24, 206, 1322, 9970, 63892, 459420, 2983714, \ \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 09:47:53, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 125$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 125$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 250$ exact terms from $n = 2$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{80} c_i\, a(n-i),$$

c_{1}	8	c_{21}	3258826705	c_{41}	311024576895	c_{61}	−11555386526707
c_{2}	52	c_{22}	−2973387241	c_{42}	−163262879798	c_{62}	4206652476337
c_{3}	−527	c_{23}	−8194146262	c_{43}	1842528177460	c_{63}	9447251803171
c_{4}	−552	c_{24}	19523134472	c_{44}	−274885928633	c_{64}	−6821481913164
c_{5}	12560	c_{25}	5607713436	c_{45}	−3005236487017	c_{65}	−4662833495886
c_{6}	−7920	c_{26}	−40873689296	c_{46}	−138289726724	c_{66}	5390611869473
c_{7}	−147919	c_{27}	23843244062	c_{47}	1668883038999	c_{67}	480291937762
c_{8}	232662	c_{28}	22390420517	c_{48}	2456017017769	c_{68}	−2547644091294
c_{9}	890118	c_{29}	−88011218639	c_{49}	1483740037317	c_{69}	829767570866
c_{10}	−2229550	c_{30}	70648129219	c_{50}	−8036508242068	c_{70}	634157227952
c_{11}	−1696482	c_{31}	178332325951	c_{51}	−2127901359355	c_{71}	−245111998652
c_{12}	9553892	c_{32}	−158366143019	c_{52}	12644141808563	c_{72}	−53739829476
c_{13}	−11958797	c_{33}	−289176114443	c_{53}	−2266382370236	c_{73}	21362320188
c_{14}	−1853450	c_{34}	148011508255	c_{54}	−7005060866344	c_{74}	−28120408
c_{15}	89426662	c_{35}	226924384926	c_{55}	5369918347133	c_{75}	9770945520
c_{16}	−179534657	c_{36}	−101382787278	c_{56}	−3671908870004	c_{76}	2147048528
c_{17}	−191684800	c_{37}	427046973892	c_{57}	−5231562047225	c_{77}	−1274905952
c_{18}	849066746	c_{38}	−60250661524	c_{58}	5136108201116	c_{78}	−8738496
c_{19}	−370826958	c_{39}	−1145037080934	c_{59}	8452368104418	c_{79}	5495040
c_{20}	−1330916663	c_{40}	343907223624	c_{60}	−2121586831231	c_{80}	8508416

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A240297 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{80-1} - \cdots - c_{80}$. Its constant term is $-c_{80} = -8508416$, which is not zero, so $q_{\min}(0) \neq 0$

and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 8508416.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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